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EECE 5644 Mini Project 2

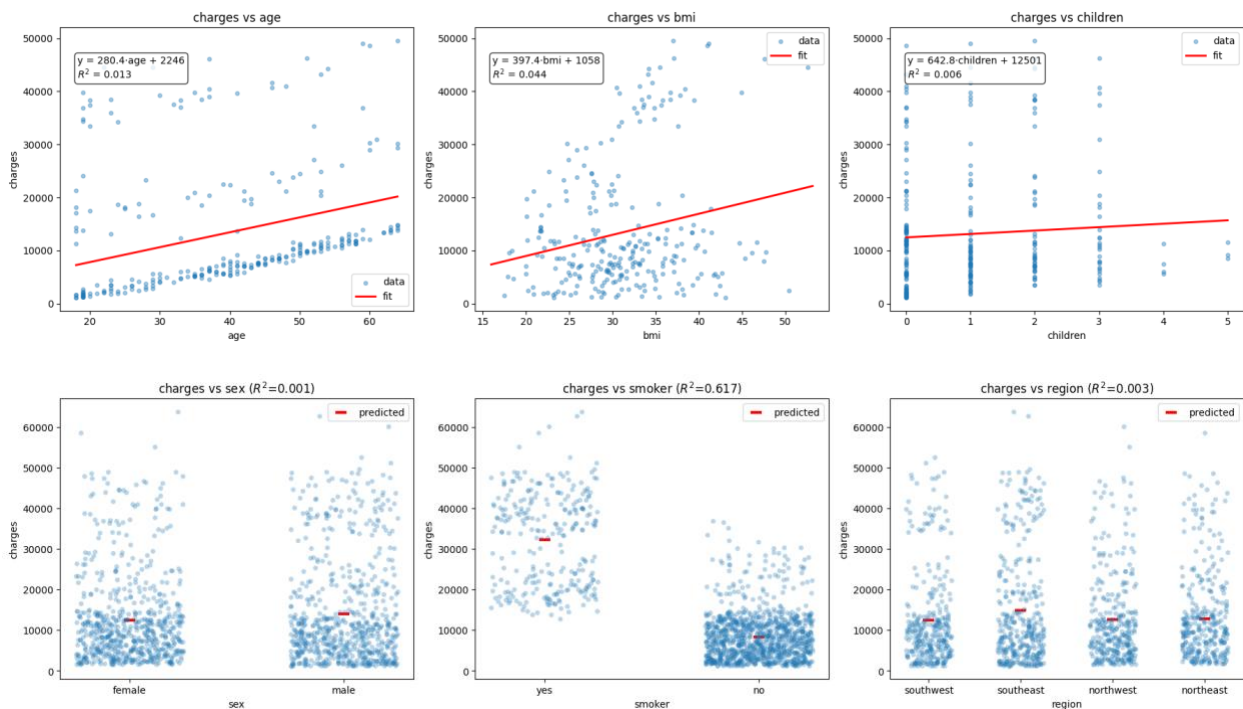
July 15, 2026

Requirement 1

A series of single variable linear regressions revealed the following results

	r2	rmse	mae
smoker	0.617377	7296.560128	5395.152670
bmi	0.043995	11533.538511	9055.705722
age	0.012829	11720.027468	9263.018844
children	0.005921	11760.960604	8883.261663
region	0.002555	11780.857130	8843.383830
sex	0.000740	11791.569459	8850.691996

The table indicates that whether a not a customer smokes has the greatest influence on their costs to the company. Whether or not a person smokes is able to explain 62% of the variance in their medical costs. The following figures display the regression lines against the actual data points for each feature.



Requirement 2

Building the Models

Simple Linear Regression

The simple linear regression model makes use of a pipeline consisting of a OneHotEncoder preprocessor and a LinearRegression model. The OneHotEncoder takes the smoking status and turns it into a binary value from the text value that was originally present. Smoking status was chosen as the feature for this model since a preliminary analysis showed that to be the best predictor of medical charges.

Multiple Linear Regression

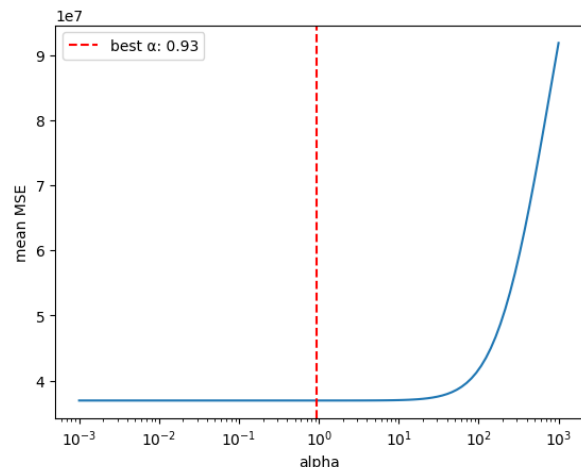
The multiple linear regression, and all other models evaluated for Requirement 2 utilize a preprocessor that uses a StandardScaler for numerical features (age, bmi, and children) and a OneHotEncoder for categorical features (sex, smoking status, region). The multiple linear regression took in all features, preprocessed them with the preprocessor, and ran them through a LinearRegression model.

Polynomial Regression

Three polynomial regressions were evaluated, degrees 2, 3, and 4. Using the same preprocessor, these pipelines implement PolynomialFeatures(degree=n), before calling LinearRegression to fit the data. This allows the LinearRegression model to fit curved features and interactions between features. It can be seen later that the degree 2 and 3 polynomials significantly outperform the degree 4 polynomial, likely due to overfitting. The degree 4 polynomial becomes too sensitive to small changes in the dataset and misses the greater trend.

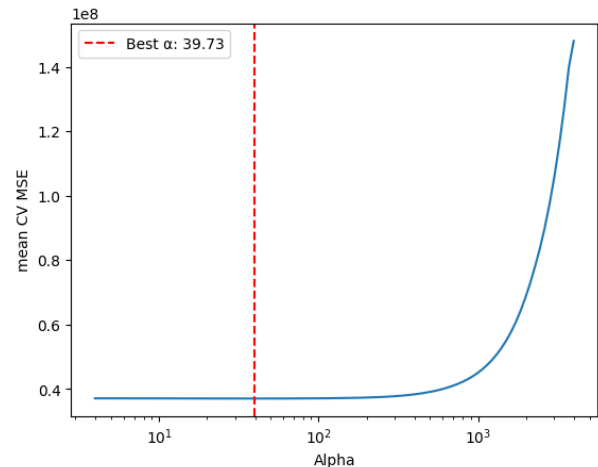
Ridge Regression

The ridge regression pipeline was implemented using the preprocessor and then calling RidgeCV, a ridge model with integrated cross-validation tuning to find the best α parameter. The automated α tuning revealed the shown plot, and $\alpha = 0.39$.



Lasso Regression

The lasso regression pipeline was similarly implemented with the LassoCV model, which automatically tunes the lasso α parameter, found to be $\alpha = 39.73$. The lasso regression eliminated the features `sex_male` and `sex_female`, as these likely had little impact on charges. It also eliminated `region_southeast` as this is colinear with the other regional features.



Support Vector Regression

Two support vector regression pipelines were created, one using a linear kernel, implemented with LinearSVR, and optimized using GridSearchCV, with `r2` score as the scoring parameter. The linear pipeline implements `squared_epsilon_insensitive` loss and a primal solver to allow the optimizer to converge. The optimized linear kernel parameters were reported as follows `C`: 0.1, ϵ : 0.0. The RBF kernel was implemented with the default parameters and optimized using GridSearchCV. The optimized RBF kernel parameters were reported as `C`: 1, ϵ : 0.1, γ : 0.1.

Decision Tree Regression

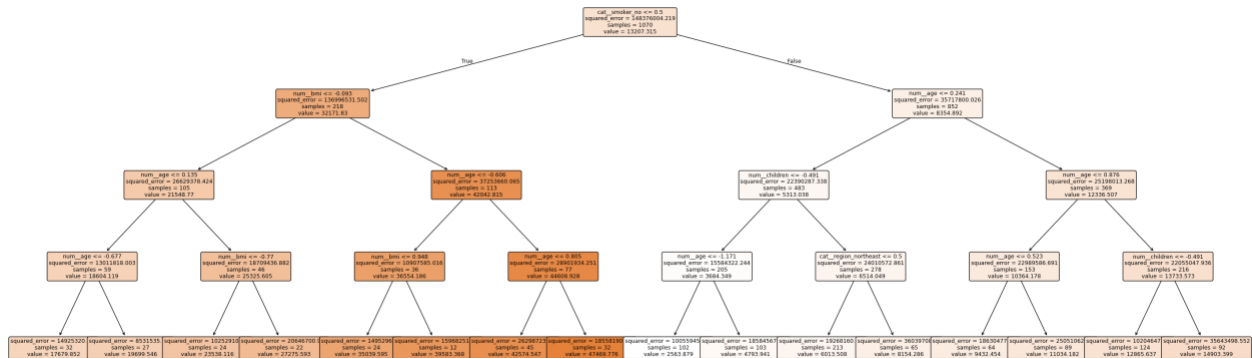
The decision tree regression pipeline was implemented with the preprocessor and `DecisionTreeRegression(max_depth=4)`, which produced the best results when compared with depths of 3 and 5.

Evaluation:

After building and optimizing each of the models, their scores were as shown in the table.

	r2	MAE	MSE	RMSE
Decision Tree Regression	0.858871	2662.684129	1.963724e+07	4431.392106
SVR RBF	0.843213	2731.605763	2.181604e+07	4670.764870
Degree 2 Polynomial	0.836526	2816.053746	2.274647e+07	4769.325455
Degree 3 Polynomial	0.829719	2974.072542	2.369361e+07	4867.608615
SVR Linear	0.729791	4427.497919	3.759802e+07	6131.722483
Ridge Regression	0.729784	4410.967276	3.759900e+07	6131.802222
Multiple Linear Regression	0.729753	4406.793913	3.760328e+07	6132.151642
Lasso Regression	0.729625	4412.744783	3.762115e+07	6133.607977
Degree 4 Polynomial	0.692798	3724.651788	4.274538e+07	6537.995253
Single Linear Regression	0.617377	5395.152670	5.323979e+07	7296.560128

The Decision Tree Regression was able to explain 86% of the variance in a customer's medical costs, closely followed by the Support Vector RBF model and degrees 2 and 3 polynomials. Based on these results, I would recommend the Decision Tree Regression model. It has the highest R^2 accuracy, lowest errors across the board, and creates an easy to interpret methodology for stakeholders. (image shown generated as req2_tree.png)



Requirement 3

Whether a customer will be expensive or not was implemented by first by categorizing costs as above or below the median cost for all customers, then by implementing a decision tree classification. This implementation was done using the preprocessor and DecisionTreeClassification with max_depth=2, found to be the same or better accuracy than higher depths.

	accuracy	precision	recall	f1
Classification Decision Tree	0.932836	0.991667	0.875	0.929688

